

Log-polyhomogeneous pseudodifferential operators on filtered manifolds and their spectral asymptotics

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In this talk I will discuss the tangent groupoid theory of pseudodifferential operators whose symbols have asymptotic expansion like

$$\sigma(x, \xi) \sim \sum_{\ell=0}^k \sum_{j=0}^{\infty} a_{j,\ell}(x, \xi) \log(|\xi|)^{\ell}$$

as $\xi \rightarrow \infty$, where $a_{j,k}(x, \xi)$ is homogeneous of order $m - j$ in ξ .

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as $\xi \rightarrow \infty$, where $a_{j,k}(x, \xi)$ is homogeneous of order $m - j$ in ξ . Log-polyhomogeneous pseudodifferential operators were introduced by Lesch in 1999.

The main theorem

Theorem

Let (M, H) be a compact filtered manifold, with homogeneous dimension Q . Let A be a k -log-polyhomogeneous pseudodifferential operator of order $-Q$. Let $\{\lambda(n, A)\}_{n=0}^{\infty}$ be the sequence of eigenvalues of A , arranged in order of decreasing absolute value. We have

$$\sum_{n=0}^N \lambda(n, A) = \frac{1}{Q(k+1)!} \text{Res}_k(A) \log(N)^k + o(\log(N)^k), \quad N \rightarrow \infty.$$

I will explain the notation and terminology later.

Section 1: What is a log-polyhomogeneous function?

Some notation

Let $V = \mathbb{R}^N$ be a graded vector space, with anisotropic dilations

$$\delta_t = \bigoplus_{k=1}^N t^{w_k}, \quad t > 0.$$

where $w_1, \dots, w_N > 0$.

Some notation

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where $w_1, \dots, w_N > 0$.

δ_t acts on distributions $K \in \mathcal{S}'(V)$ by pushforward.

$$((\delta_t)^* K, f) = (K, f \circ \delta_t^{-1}), \quad t > 0, f \in \mathcal{S}(V).$$

Homogeneous and almost-homogeneous functions

We say that $K \in \mathcal{S}'(V)$ is *homogeneous* of degree $m \in \mathbb{C}$ if

$$(\delta_t)^* K = t^m K, \quad t > 0$$

and *almost homogeneous* of degree $m \in \mathbb{C}$ if

$$(\delta_t)^* K - t^m K \in \mathcal{S}(V), \quad t > 0.$$

Infinitesimal forms of homogeneity

Let $Zf := \frac{d}{dt}|_{t=1} f \circ \delta_t$ be the generator of the dilations δ_t , i.e.

$$Z := \sum_{j=1}^N w_j x_j \frac{\partial}{\partial x_j}$$

Theorem (Euler, 18th century)

A distribution $K \in \mathcal{S}'(V)$ is homogeneous of degree m if and only if

$$(Z - m)K = 0.$$

Similarly...

Theorem

A distribution $K \in S'(V)$ is almost-homogeneous of degree m if and only if

$$(Z - m)K \in \mathcal{S}(V).$$

Taylor's lemma

Taylor proved that an almost-homogeneous function is almost a homogeneous function.

Lemma (Taylor(?), 1984)

Let $K \in C^\infty(V) \cap \mathcal{S}'(V)$. If K is almost homogeneous of degree m , then there exists a homogeneous function $H \in C^\infty(V \setminus \{0\})$ such that for any $\psi \in C_c^\infty(V)$ with $\psi = 1$ near zero, we have

$$K - (1 - \psi)H \in \mathcal{S}(V).$$

Obviously we have

$$H(\xi) = \lim_{t \rightarrow \infty} t^{-m} K(\delta_t \xi), \quad x \in V \setminus \{0\}.$$

The theorem more-or-less amounts to the identity

$$t^{-m} K(\delta_t \xi) - K(\xi) = \int_1^t s^{-m-1} ((Z - m)K)(\delta_s \xi) ds, \quad t > 0, \xi \in \mathbb{R}^N.$$

Polyhomogeneous functions

A function $K \in C^\infty(V)$ is *polyhomogeneous* if there exist homogeneous functions $\{K_\ell\}_{\ell=0}^\infty$ of degrees $\{m - \ell\}_{\ell=0}^\infty$ such that

$$K(\xi) \sim \sum_{\ell=0}^{\infty} K_\ell(\xi), \quad |\xi| \rightarrow \infty$$

in the usual sense.

Theorem (Debord-Skandalis, 2014)

Let $\tilde{\delta}_t(\xi, h) := (\delta_t \xi, th)$. A smooth function K on V is polyhomogeneous of order m if and only if there exists an almost homogeneous function $\tilde{K}$ on $V \times \mathbb{R}$ such that

$$K(\xi) = \tilde{K}(\xi, 1), \quad \xi \in \mathbb{R}^N.$$

Moreover, $\tilde{K}(\xi, 0)$ is an almost-homogeneous function whose homogeneous part is the principal term K_0 .

Log-homogeneous functions

Let $|\cdot|$ be a δ -homogeneous norm on V , i.e. $|\delta_t \xi| = t|\xi|$. Functions of the form $L(\xi) = \log(|\xi|)$ are special because $L \circ \delta_t = \log(t) + L$, or in other words

$$ZL = 1.$$

Log-homogeneous functions

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$$ZL = 1.$$

We will say that a function $K \in C^\infty(V \setminus \{0\})$ is log-homogeneous of order (m, k) if there exist $H_0, \dots, H_k$, homogeneous functions of order m , such that

$$K(\xi) = \sum_{\ell=0}^k H_\ell(\xi) \log(|\xi|)^\ell.$$

Log-homogeneous functions

Lemma

A function $K \in C^\infty(V \setminus \{0\})$ is log-homogeneous of order (m, k) if and only

$$(Z - m)^k K = 0.$$

Proof: Induction on k .

Almost-log-homogeneous functions

By analogy with almost-homogeneous functions, say that $K \in C^\infty(V)$ is almost log-homogeneous of order (m, k) if

$$(Z - m)^{k+1}K \in \mathcal{S}(V)$$

Lemma (Logarithmic Taylor lemma)

Let $K \in C^\infty(V) \cap \mathcal{S}'(V)$. If K is almost homogeneous of degree m , then there exists a log-homogeneous function $H \in C^\infty(V \setminus \{0\})$ such that for any $\psi \in C_c^\infty(V)$ with $\psi = 1$ near zero, we have

$$K - (1 - \psi)H \in \mathcal{S}(V).$$

log-polyhomogeneous functions

Say that a function $K \in C^\infty(V)$ is log-polyhomogeneous of order (m, k) if there exist homogeneous functions $\{K_{\ell,j}\}_{j=0}^\infty$ such that

$$K(\xi) \sim \sum_{\ell=0}^k \sum_{j=0}^{\infty} \log(|\xi|)^j K_{\ell,j}(\xi), \quad |\xi| \rightarrow \infty.$$

Theorem

Let $\tilde{\delta}_t(\xi, h) := (\delta_t \xi, th)$. A smooth function K on V is log-polyhomogeneous of order (m, k) if and only if there exists an almost homogeneous function $\tilde{K}$ on $V \times \mathbb{R}$ such that

$$K(\xi) = \tilde{K}(\xi, 1), \quad \xi \in \mathbb{R}^N.$$

Moreover, $\tilde{K}(\xi, 0)$ is an almost log-homogeneous function whose log-homogeneous part is the principal term $\xi \rightarrow \log(|\xi|)^k K_{k,0}(\xi)$.

Section 2: What is an operator with log-polyhomogeneous symbol?

The H -tangent groupoid

Let M be a manifold, and let

$$0 = H^0 < H^1 < \dots < H^N = TM$$

be a Lie filtration of the tangent bundle. We have an osculating bundle $T_H M$ of graded nilpotent Lie groups and an H -tangent groupoid

$$\mathbb{T}_H M := (T_H M \times \{0\}) \sqcup (M \times M \times \mathbb{R}^\times).$$

which is equipped with the correct topology.

Reminder on notation

- (M, H) is a filtered manifold.
- $Q = \sum_{k=1}^{\infty} k(\text{rank}(H^k) - \text{rank}(H^{k-1}))$ is its homogeneous dimension.
- $T_H M$ is the osculating groupoid, with range map

$$r : T_H M \rightarrow M$$

- $\mathbb{T}_H M$ is the H -tangent groupoid, with range map

$$r : \mathbb{T}_H M \rightarrow \mathbb{R} \times M.$$

- $\mathcal{E}'_r(G)$ is the space of r -fibred distributions

$$C^\infty(G) \rightarrow C^\infty(G^{(0)})$$

- $\Omega_r := \Omega_{\ker(dr)}$ is the bundle of densities transversal to the r -fibres.
- $C_p^\infty(G, \Omega_r)$ is the space of r and s properly supported r -transversal densities.

The H -tangent groupoid

The zoom action on $\mathbb{T}_H M$ is

$$\alpha_t(x, y, h) = (x, y, t^{-1}h), \quad \alpha_t((x, z), 0) = ((x, \delta_t z), 0).$$

The generator of the zoom action

Let Z be the vector field on $\mathbb{T}_H M$ which generates the zoom action, i.e.

$$Zf := \left. \frac{d}{dt} \right|_{t=1} f \circ \alpha_t.$$

The Lie derivative by Z acts on r -fibred distributions, i.e.

$$\mathcal{L}_Z \mathbb{P} := \left. \frac{d}{dt} \right|_{t=1} (\alpha_t)_* \mathbb{P}.$$

The van Erp-Yuncken calculus

Definition (van Erp-Yuncken (2019))

A pseudodifferential family $\mathbb{P}$ of order m is an r -fibred distribution on $\mathbb{T}_H M$ such that

$$(\mathcal{L}_Z - m)\mathbb{P} \in C_p^\infty(\mathbb{T}_H M, \Omega_r).$$

A pseudodifferential operator P of order m is defined by its Schwartz kernel being of the form $\mathbb{P}_{h=1}$ for some pseudodifferential family $\mathbb{P}$.

Moreover, if $\mathbb{P}_1 = P$, then $\mathbb{P}_0$ is an approximately homogeneous distribution on $T_H M$ which represents the principal (co)symbol of P .

The log-polyhomogeneous van Erp-Yuncken calculus

The natural thing to do is to define k -log-polyhomogeneous operators in the same way.

Definition

A k -log-polyhomogeneous family is an r -fibred distribution $\mathbb{P} \in \mathcal{E}'_r(\mathbb{T}_H M)$ such that

$$(\mathcal{L}_Z - m)^{k+1} \mathbb{P} \in C_p^\infty(\mathbb{T}_H M, \Omega_r).$$

A pseudodifferential operator P of order (m, k) is defined by its Schwartz kernel being of the form $\mathbb{P}_{h=1}$ for some pseudodifferential family $\mathbb{P}$.

Denote the set of pseudodifferential operators with k -log-polyhomogeneous order (m, k) as $\Psi_H^{m,k}(M)$.

Yes, it is the right definition

Theorem (M. 2026)

If $H = \text{triv.}$ is the trivial filtration, then $\Psi_H^{m,k}(M)$ is the operators which locally have symbol function that is log-polyhomogeneous of order (m, k) .

Some obvious facts

It is pretty straightforward to check that $\bigcup_{m,k} \psi_H^{m,k}(M)$ is bi-graded by m and k , i.e.

$$\psi_H^{m_0,k_0}(M) \cdot \psi_H^{m_1,k_1}(M) \subseteq \psi_H^{m_0+m_1,k_0+k_1}(M).$$

We also have closure under the adjoint operation, etc. etc.

A less-than-obvious fact

The symbol mapping goes like

$$0 \rightarrow \Psi_H^{m-1,k}(M) \rightarrow \Psi_H^{m,k}(M) \rightarrow \Sigma_H^{m,k}(M) \rightarrow 0.$$

where the symbol class $\Sigma_H^{m,k}(M)$ has a non-obvious definition.

Definition

Let

$$\Sigma_H^{m,k}(M) := \{T \in \mathcal{E}'_r(T_H M) : (\mathcal{L}_Z - m)^{k+1} T \in C_p^\infty(T_H M, \Omega_r)\}, \quad k \geq 0$$

and

$$\Sigma_H^{m,k}(M) := \Sigma_H^{m,k}(M) / \Sigma_H^{m,k-1}(M), \quad k \geq 1.$$

Section 3: The residue and the trace

Back to functions

Again let V be a vector space with a group of dilations $\{\delta_t\}_{t>0}$. Let $S_{\text{phg}}^m(V)$ be the set of polyhomogeneous smooth functions of order m , i.e.

$$S_{\text{phg}}^m(V) = \{f \in C^\infty(V) : f \sim \sum_{j=0}^{\infty} f_j, f_j(\delta_t \xi) = t^m f_j(\xi)\}.$$

There are two important functionals on $S_{\text{phg}}^m(V)$, depending on the value of m . They are the *canonical integral*,

$$\text{Int} : S_{\text{phg}}^m(V) \rightarrow \mathbb{C}, \quad m \in \mathbb{C} \setminus \{-Q, -Q+1, -Q+2, \dots\}$$

and the *residue*

$$\text{Res} : S_{\text{phg}}^m(V) \rightarrow \mathbb{C}, \quad m \in \mathbb{Z}.$$

Debord-Skandalis description of the canonical integral and the residue

Let $K \in S_{\text{phg}}^m(V)$. By Debord-Skandalis' characterisation, there exists an almost-homogeneous function $\tilde{K}$ on $V \times \mathbb{R}$ such that

$$K(\xi) = \tilde{K}(\xi, 1).$$

Let

$$a(\xi, h) := \left. \frac{d}{dt} \right|_{t=1} (t^{-m} \tilde{K} \circ \tilde{\delta}_t) = (\tilde{Z} - m) \tilde{K}(\xi, h).$$

and $r(h) := \int_V a(\xi, h) d\xi$. We have

$$\text{Res}(K) = r(0)$$

and for sufficiently large n ,

$$\begin{aligned} \text{Int}(K) &= \sum_{j=0}^n \frac{1}{j!} \frac{r^{(j)}(0)}{j - m - Q} \\ &\quad + \frac{1}{(n-1)!} \int_0^1 r^{(n)}(u) u^{n-m-Q-1} B(1-u; n, m+Q-n+1) du \end{aligned}$$

Section 4: Spectral theory and the trace theorem

Let $\mathcal{H}$ be a Hilbert space, and $\mathcal{K}(\mathcal{H})$ is the ideal of compact operators on $\mathcal{H}$. For $T \in \mathcal{K}(\mathcal{H})$, let $\lambda(T) = \{\lambda(n, T)\}_{n=0}^{\infty}$ be a sequence of eigenvalues of T .

- ① We arrange the sequence such that $|\lambda(n, T)|$ is non-increasing. This is ambiguous, but ultimately unimportant.
- ② If T has finitely many eigenvalues (including none at all, this can happen in the non-normal case!) then $\lambda(T)$ should be filled out with zeros.

Weyl's inequality

$\lambda(n, |T|)$ is the $(n+1)$ th singular value of T . Note that $|\lambda(n, T)|$ and $\lambda(n, |T|)$ are very different in general. The best we can say is that

$$\prod_{n=0}^N |\lambda(n, T)| \leq \prod_{n=0}^N \lambda(n, |T|), \quad N \geq 0.$$

(This is Weyl's inequality).

Definition

Let $p, k \geq 0$. Define

$$\mathcal{E}_{p,k}(\mathcal{H}) := \{T \in \mathcal{K}(\mathcal{H}) : \lambda(n, |T|) = O\left(\frac{\log(n)^k}{n^{1/p}}\right)\}.$$

Elementary eigenvalue estimates ensure that

$$\mathcal{E}_{p,k} \cdot \mathcal{E}_{q,\ell} \subseteq \mathcal{E}_{r,k+\ell}, \quad r^{-1} = p^{-1} + q^{-1}.$$

Log-polyhomogeneous operators are weak-trace-class

Theorem (M. (2026))

Let (M, H) be a closed filtered manifold. For $m > 0$, we have

$$\Psi_H^{-m,k}(M) \subset \mathcal{E}_{\frac{Q}{m},k}(L_2(M)).$$

Eigenvalue estimates for weak-trace-class operators

Weyl's inequality implies that if $A \in \Psi_H^{-m,k}(M)$, then

$$\lambda(n, A) = O(n^{-\frac{Q}{m}} \log(n)^k), \quad n \rightarrow \infty.$$

Probably we can do better than this, and get

$$\lambda(n, A) \sim c(A) n^{-\frac{Q}{m}} \log(n)^k$$

for some $c(A)$, but this is out of reach at the moment.

Asymptotics for sums of eigenvalues

The best I can do at the moment concerns $m = Q$ and spectral averages:

Theorem (M. (2026))

Let $A \in \Psi_H^{-Q,k}(M)$. We have

$$\sum_{n=0}^N \lambda(n, A) = \frac{1}{(k+1)!Q} \operatorname{Res}_k(A) \log(N)^{k+1} + o(\log(N)^{k+1}).$$

Thank you for listening!